Rational Decision Making

Decision Analysis Examples

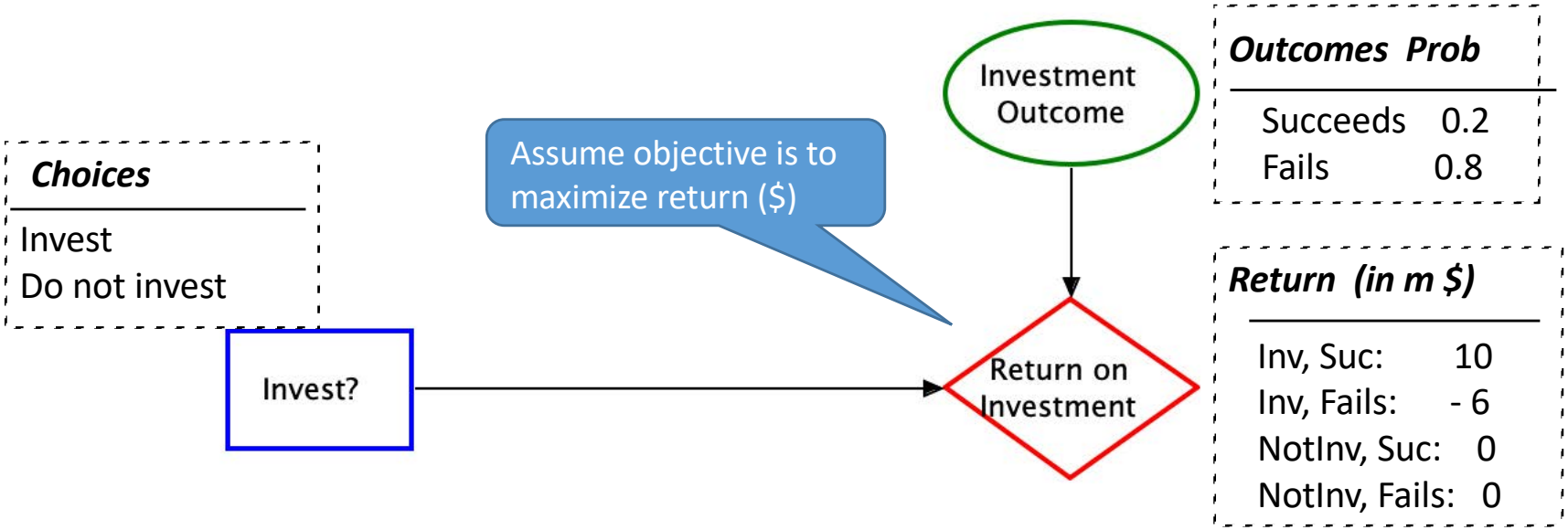
CS3263

Foundations of Artificial Intelligence

Investment Decision

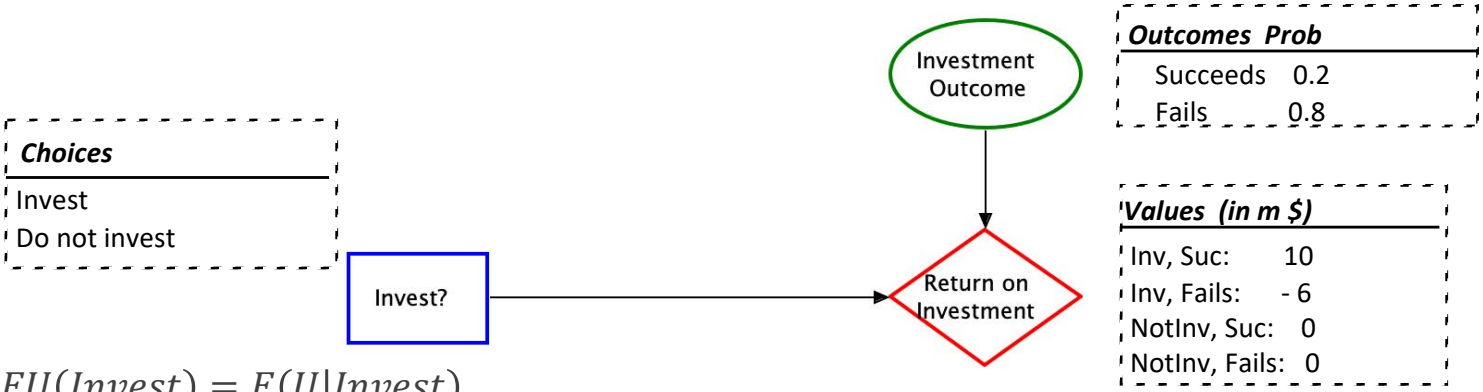
Example: The Basic Risky Decision

- No information is available when decision is made



Example: Cryptocurrency Investment Problem

Solving an Influence Diagram



$$EU(Invest) = E(U|Invest)$$

$$= P(Suc|Invest)U(Inv, Suc) + P(Fails|Invest)U(Inv, Fails)$$

$$= 0.2\times10 + 0.8\times-6 = -2.8$$

$$EU(a) = \sum_{s'}\sum_{s} P(s)P(s'|s,a)U(s')$$

$$EU(Do\ Not\ Invest) = E(U|Do\ Not\ Invest)$$

$$= P(Suc|Do\ Not\ Invest)U(NotInv, Suc) + P(Fails|Do\ Not\ Invest)U(NotInv, Fails)$$

$$= 0.2\times0 + 0.8\times0 = 0$$

$$EU(Invest?) = \max\big(EU(Invest), EU(Do\ Not\ Invest)\big) = \max(-2.8, 0) = 0$$

$$\text{“}Optimal\text{”}\ action = \underset{a}{\operatorname{argmax}}\, EU(a) = Do\ Not\ Invest$$

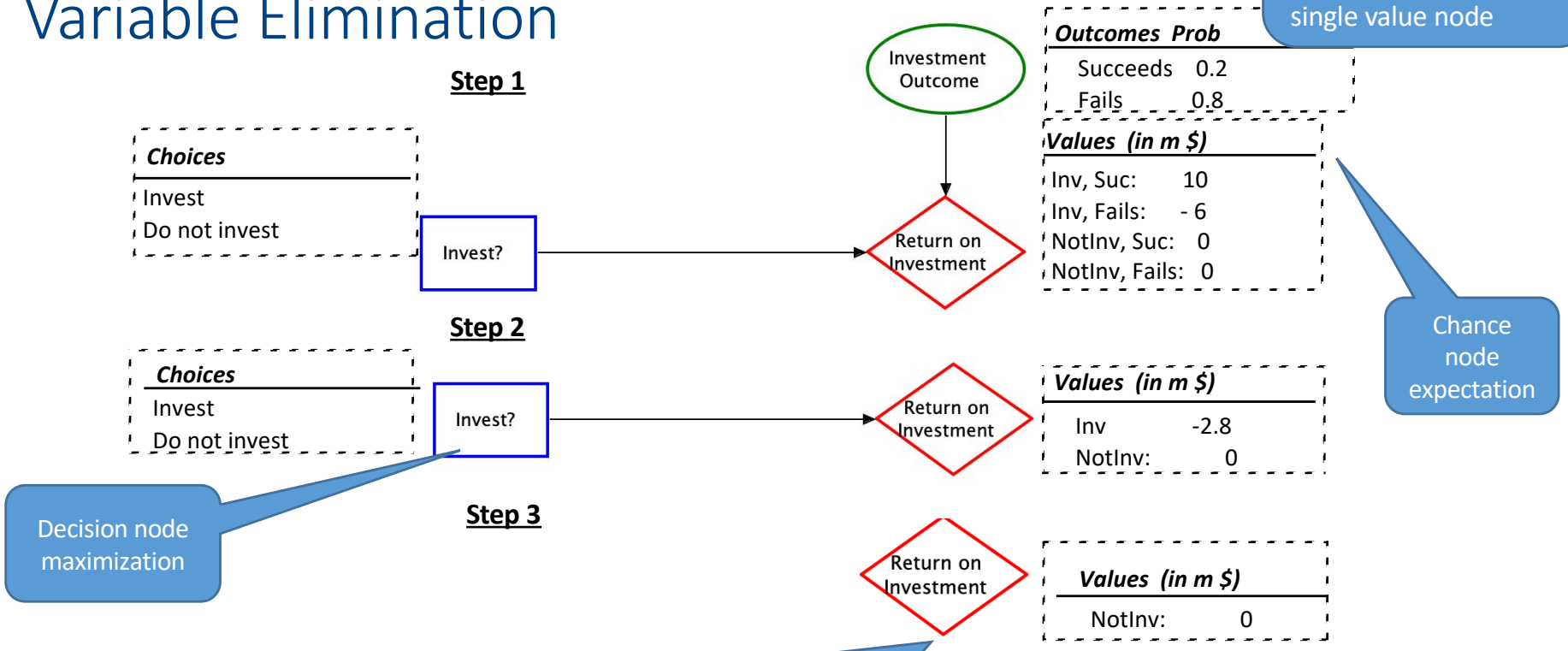
Solving an Influence Diagram:
Probabilistic Inference

- Set evidence variables to the observed states
- For each possible value of the decision node:
  - Set the decision node to that value
  - Calculate posterior probability of the parents of the utility using probabilistic inference algorithm
  - Calculate resulting utility for the action
- Return the action with the highest utility

Use efficient Bayesian Network inference algorithms

Solving an Influence Diagram:
Variable Elimination

Solution obtained by reducing diagram, while recording optimal choices, into single value node



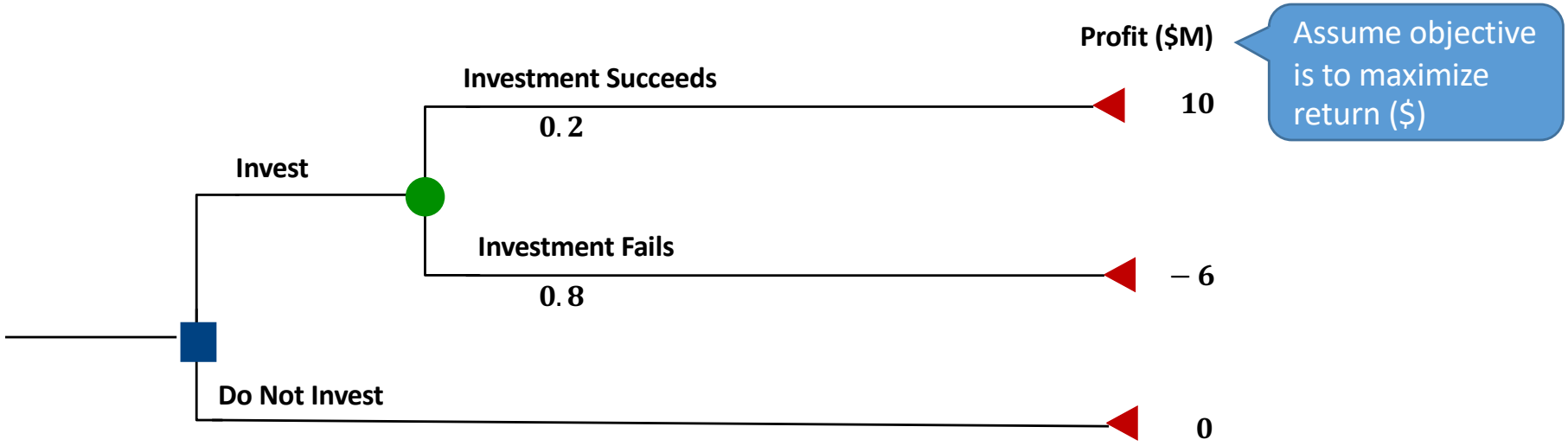
Chance node expectation

Decision node maximization

Final value node contains expected maximum value achievable with optimal decisions

Example: The Basic Risky Decision

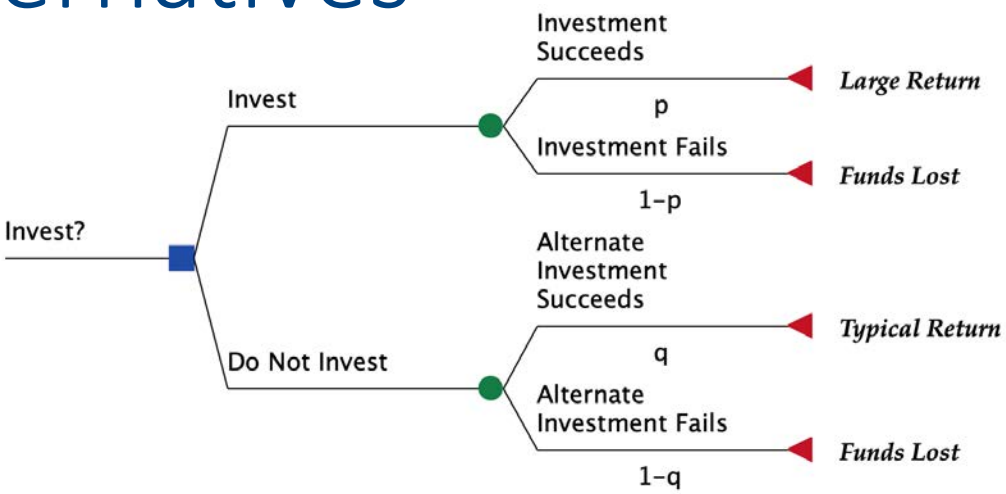
- No information is available when decision is made



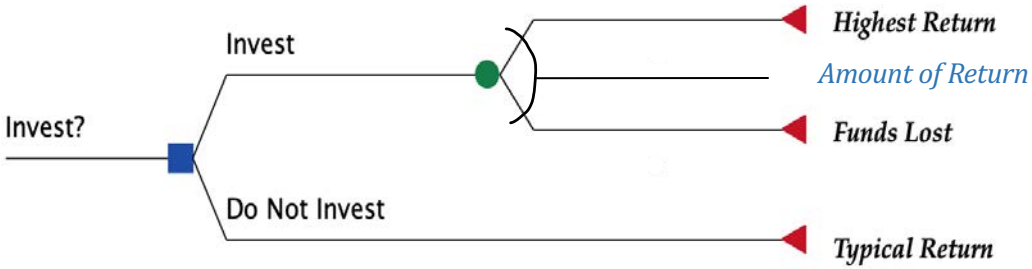
Example: Cryptocurrency Investment Problem

Decision Tree Alternatives

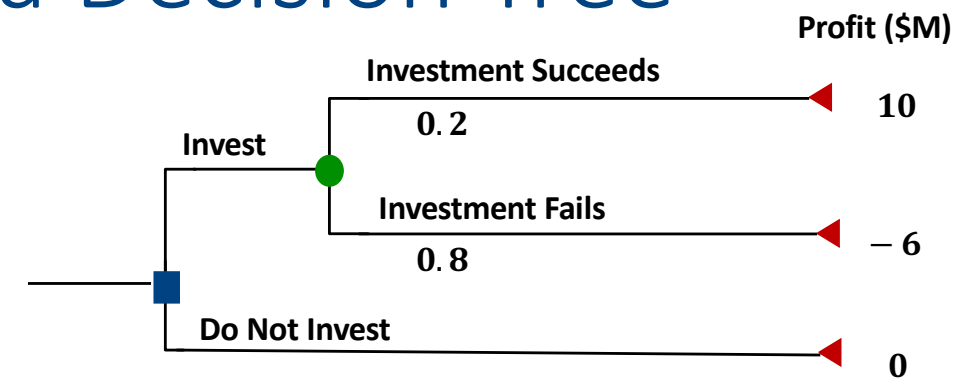
- Double Risky Decision
  - e.g., if the investor decides not to invest, she may lose the money in the stock market or the less risky investment



- Range of Risk Decision
  - e.g., Return of the original risky decision may take a range of values



Solving a Decision Tree



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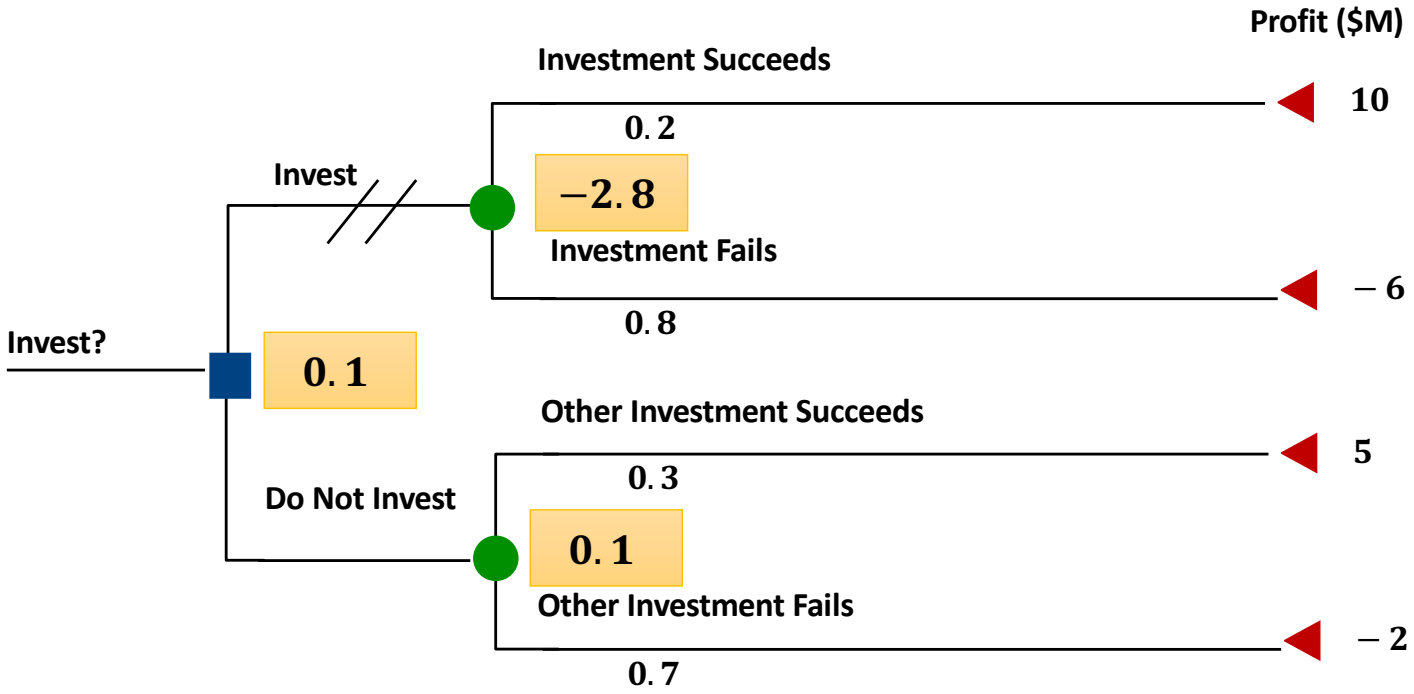
Solving a Decision Tree:
Rollback OR Simulation

- Folding or rolling back the tree:
  - Begin at end points of branches on far right-hand side and move to left
  - Chance value expectation
    - Calculate expected values when encountering a chance node; OR
  - Decision value maximization
    - Choose the branch with the highest value or expected value when encountering a decision node
  - Record value accumulated and alternative selected for each decision node along optimal path
- OR Simulation – Roll forward or sampling:
  - Monte Carlo Simulation
  - Cohort simulation
- Choose among risky alternatives
  - Pick alternative with the highest utility

Solving a Decision Tree (Rollback)

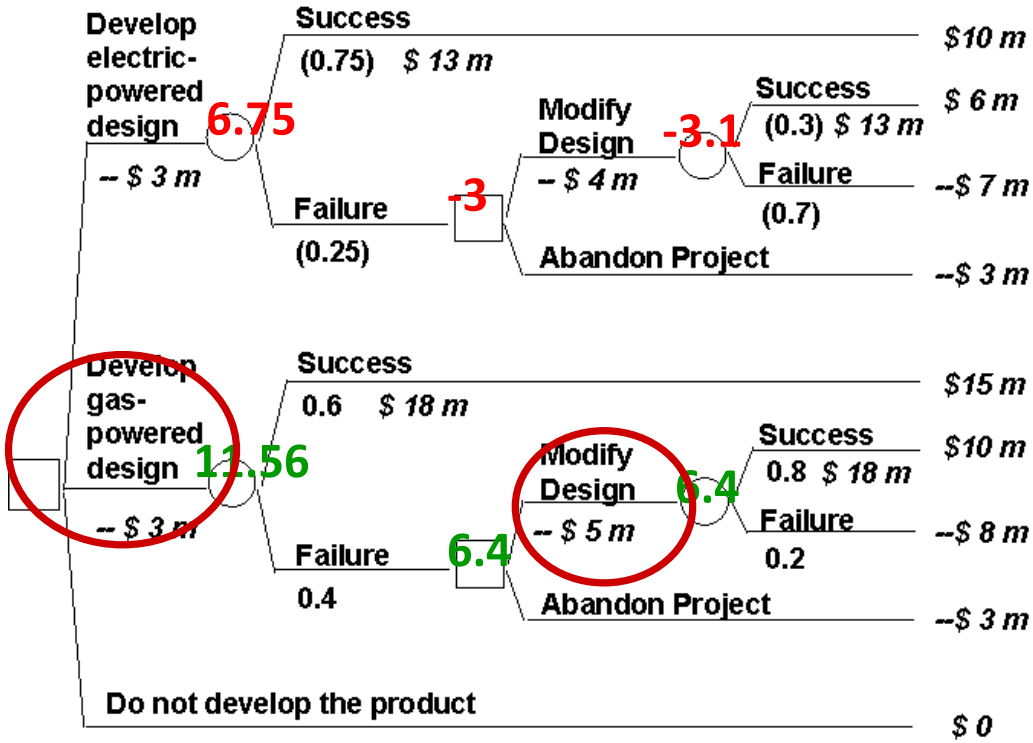


Product Development

Example: Product Development

- Problem Definition:
  - Decision on whether to develop a new type of food processor
  - Two alternative power sources may be used: electricity and gas
  - The designs have different development costs
  - Either design may succeed or fail with some uncertainty
  - Only one design can be pursued initially due to resource constraints
  - If either design fails, company would still consider modifying design, with potential more investment, and would still not guarantee success
- Two stage decision problem:
  - Decision between designs and not to develop the new processor at all
  - Decision on whether the design should be modified

Example: The Rollback Algorithm



Martian Adventures

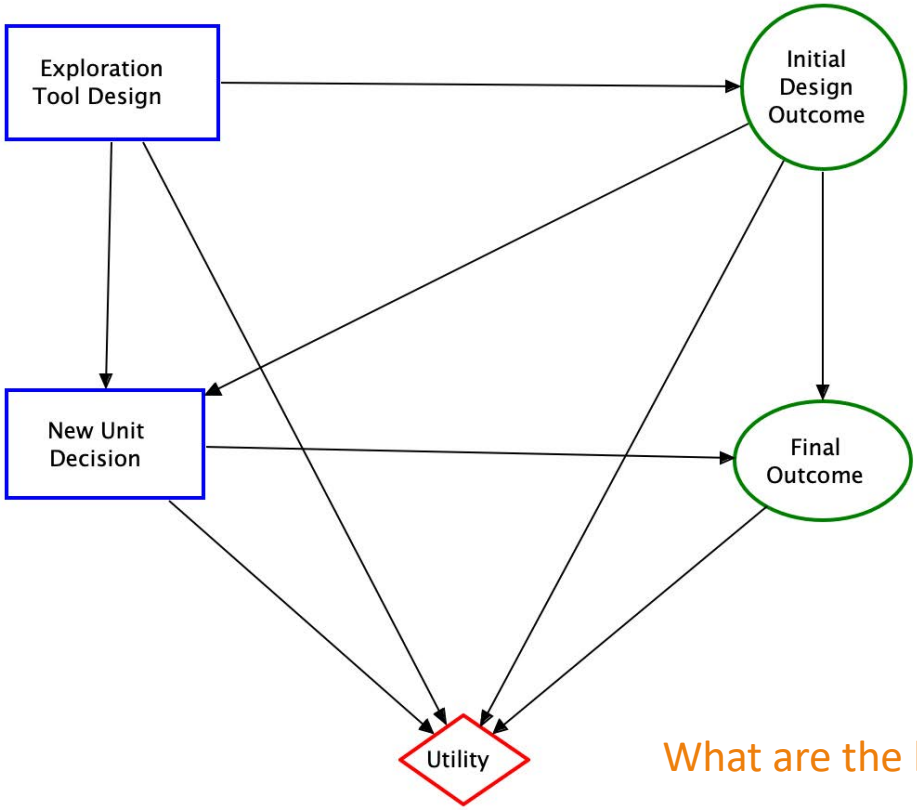
Example: Martian Adventures Cont …

- Scenario:
  - Mars Base is considering deploying a new tool for Persy the Mars Rover to collect and analyze Martian rock samples:
  - Two different power sources: Solar Power and Battery Power.
  - Both power designs have different energy consumption rates and development time.
  - Success rate for either design working efficiently in the harsh Martian environment is uncertain.
  - Only one design can be deployed at a time due to payload constraints on the resupply ships.
  - If either design doesn't work optimally once deployed, a new, modified version may be sent in the next resupply mission, but this means waiting longer and it still doesn’t guarantee success.
- Two stage decision problem:
  - Decide on which power design to deploy: Solar, Battery, or neither (if risk is too high).
  - If the tool fails, decide whether to send a modified version or to abort the mission objective.

Example: A Decision Model (1)

Which power design to use?

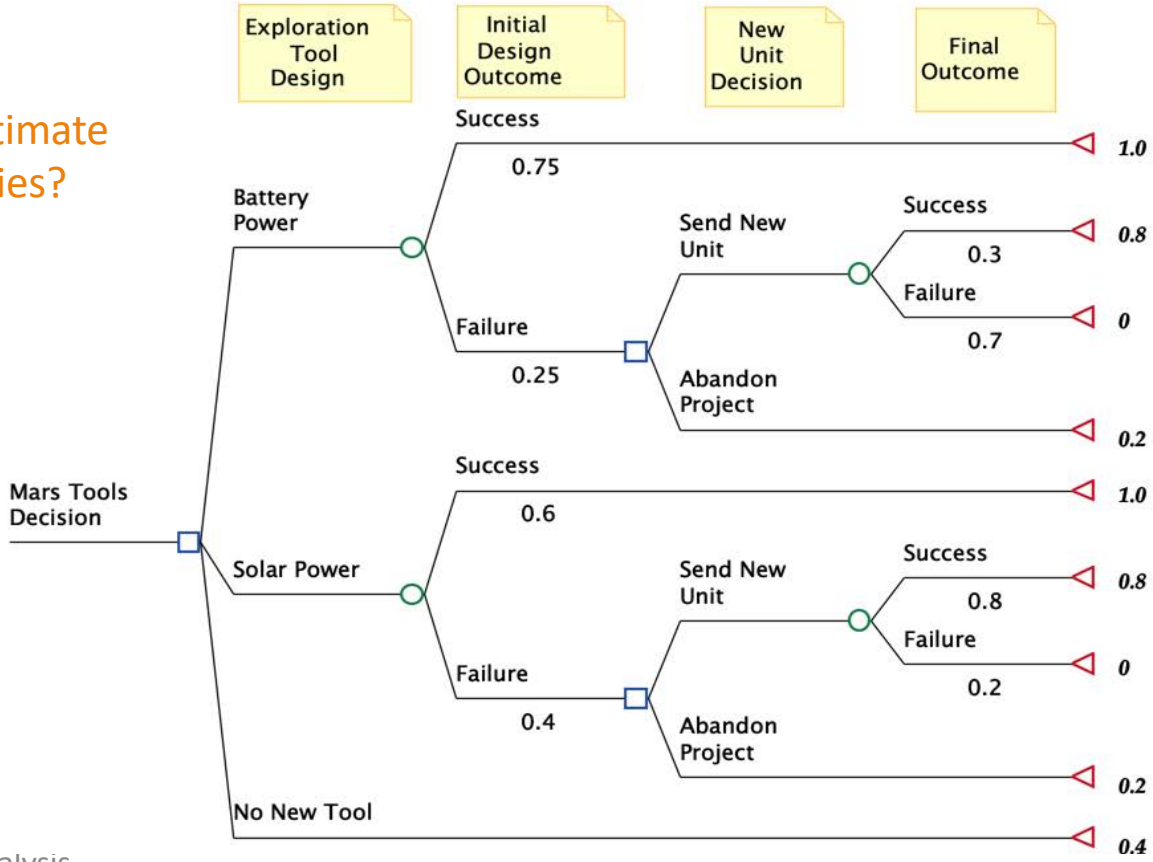
If tool fails, should a new version be sent?

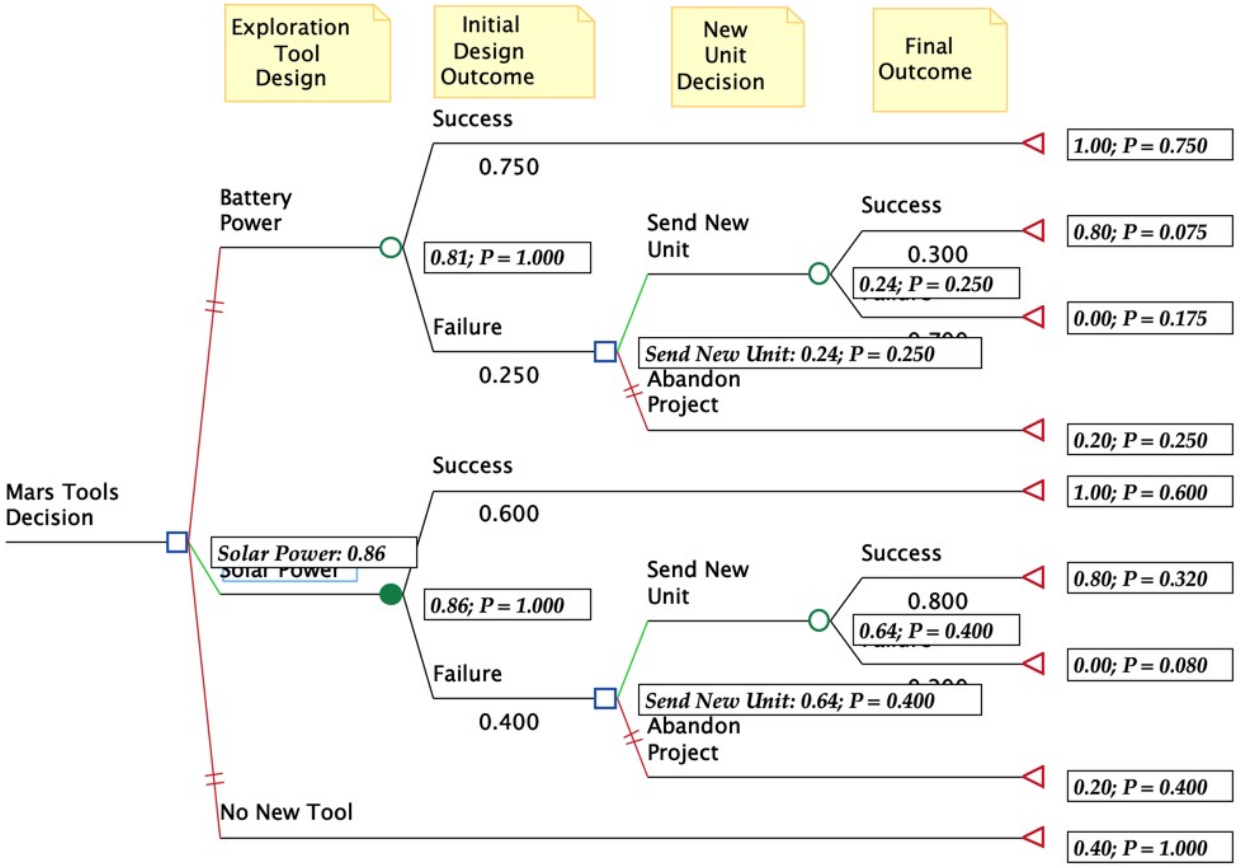


What are the benefits or costs?

Example: A Decision Model (2)

How to estimate uncertainties?



How to estimate utilities?

Example: The Rollback Algorithm